

INFORMATION FOR FIRST AND SECOND RESPONDERS
EMERGENCY RESPONSE GUIDE



Hybrid Electric Propulsion



700V Lithium-Ion Battery
Storage System

Vehicle Name/Model:
Pierce® Volterra™ Electric Fire Truck

Vehicle Type/Designation
Sub-Category:
Emergency Vehicle



 Oshkosh Corporation 1917 Four Wheel Drive Oshkosh, WI 54902	Document ID	Revision	Revision Date	Page
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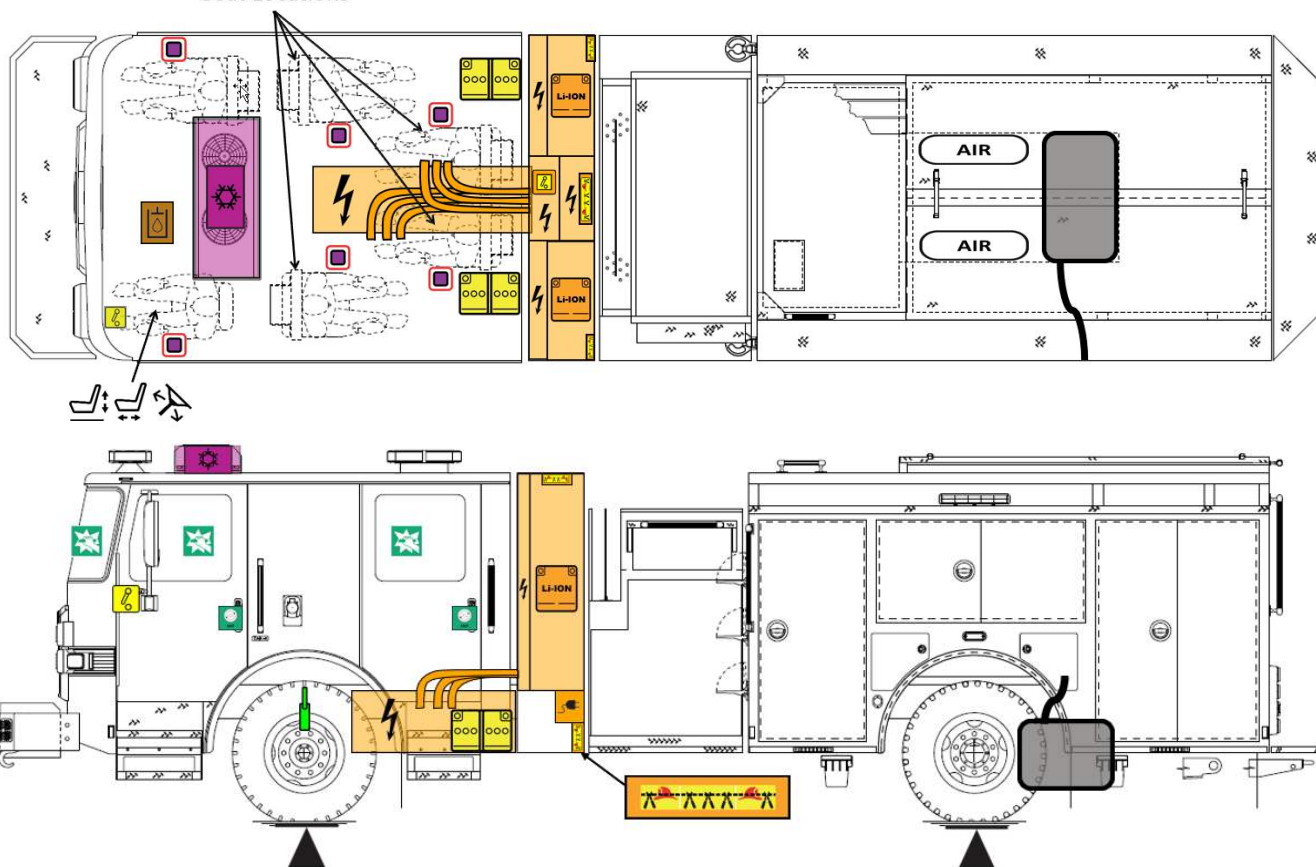


Pierce® Volterra™ Electric Fire Truck

2023 to Present



Optional Crew Cab
Seat Locations



	Hybrid Electric Propulsion		High Voltage Battery 700V LI ION		Low Voltage Battery 12V		High Voltage Disconnect		Low Voltage Disconnect		High Voltage Disconnect Cut Loop
	High Voltage Component		High Voltage Power Cable		High Voltage Charging Inlet		Gas Strut/ Preloaded Spring		Air-conditioning component		Hydraulic Fluid Reservoir
	Diesel Fuel Tank		Compressed Air Tank		Lifting Points		Entry/Exit Points		Break Points		Seat Belt Pretensioner
	Seat Height Adjustment		Seat Longitudinal Adjustment		Steering Wheel Tilt Control	Note: These images show information for one variant of this product. The Cab, Pump House and Rear Body configuration can vary depending on the variant of the product.					



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1. Identification / Recognition

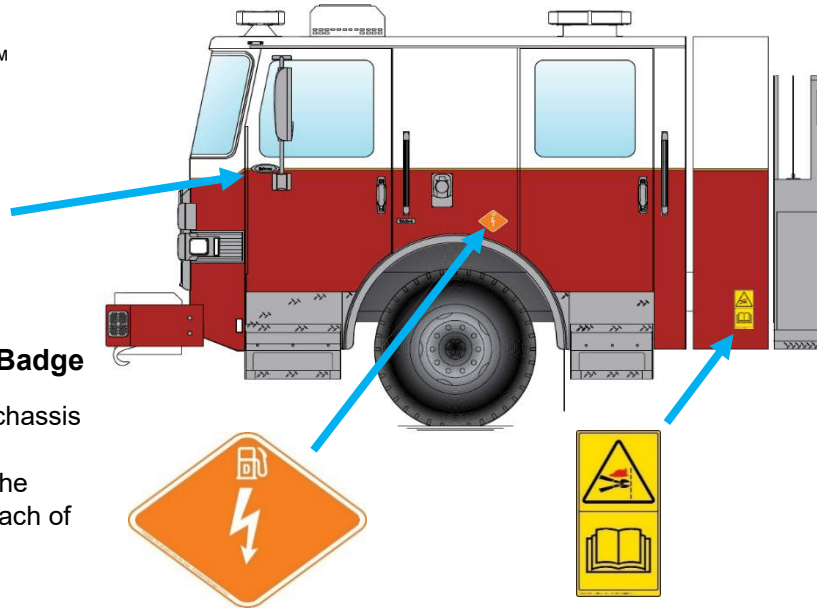
CAB SIDES

Pierce® Enforcer Volterra™



Pierce® Velocity Volterra™ Badge

The electric drivetrain equipped chassis can be identified by the exterior Volterra™ badge located under the chassis identification badge on each of the left and right front cab doors.



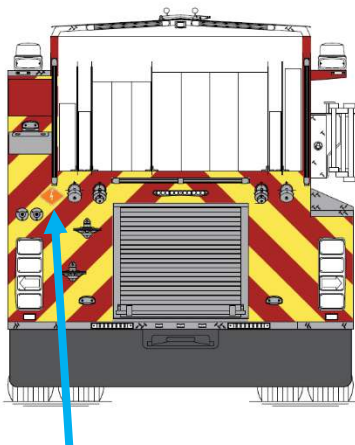
FAMA60 Hybrid Electric Propulsion Energy Identification Sign

This sign shall be located on each side of the cab and at the rear of the apparatus.

FAMA58 First Responder Loop Safety Sign

There are 3 first responder cut loop access doors labeled with FAMA58 signs on the high voltage battery rack: left side, right side and top of the rack.

REAR OF TRUCK



FAMA60 Identification sign located on the left side rear bulkhead.

INSIDE CAB / VIN



There is a Volterra™ Electric Vehicle Drivetrain label inside the cab near the Federal Motor Vehicle Safety Standard (FMVSS) information decal. This label describes the location of the High Voltage Interlock Cut Loops.



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2. Immobilization / Stabilization / Lifting



WARNING: Vehicle noise may be reduced in some operation modes. Failure to shutdown vehicle before immobilization may result in death, severe injury, or property damage.

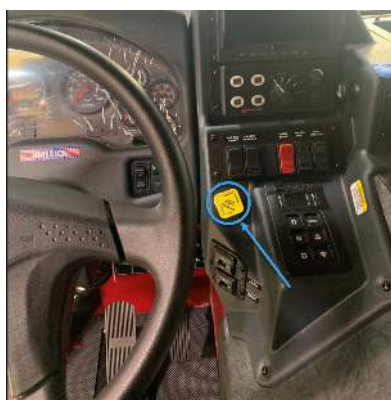


WARNING: Unintended Movement Hazard. Failure to follow shutdown procedures could result in unintended vehicle movement. Always approach the vehicle from the sides to stay out of the potential travel path and properly chock the wheels. Reduced vehicle noise may make it difficult to determine if the truck is energized. Failure to comply may result in serious personal injury or death.



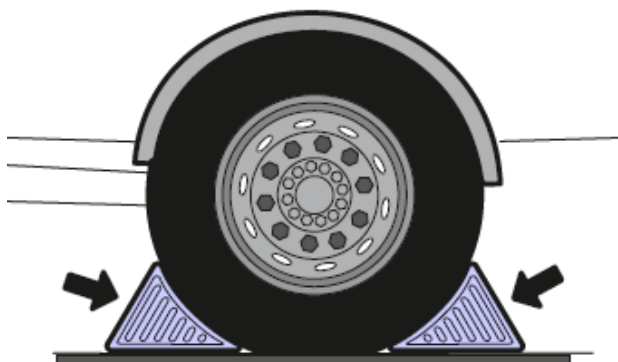
WARNING: Lifting Hazard. Do not use high voltage battery or any high voltage components to lift or stabilize vehicle. Use only designated lifting points. Failure to comply may result in serious personal injury or death.

Park Brake



Pull out to apply brake; push in to release. Applying the parking brake automatically shifts the vehicle to neutral (N).

Wheel Chocks



Properly Chock Wheels to prevent vehicle movement.



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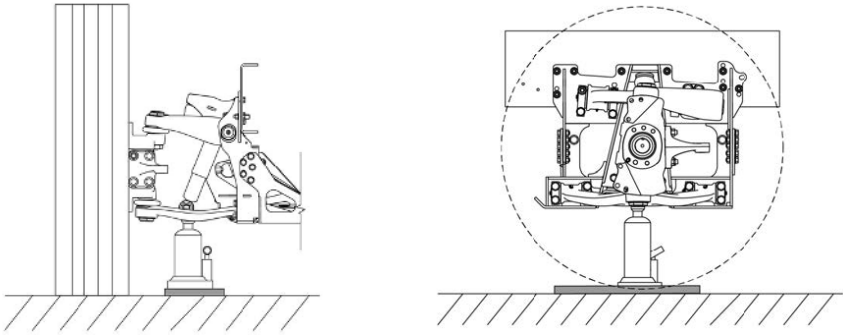
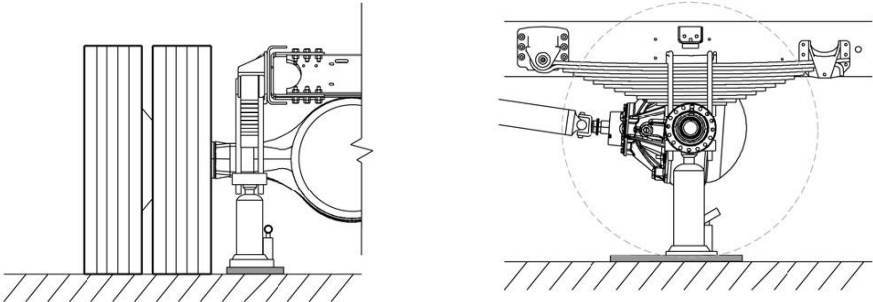
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Front Axle Lifting Location 	Rear View Side View  Lift under the lower control arm so it supports the lower control arm under the shock mount.
Rear Axle Lifting Location 	Rear View Side View  Lift under the axle below the spring mount

3. Disable Direct Hazards / Safety Regulations



WARNING: Electrical Shock Hazard. Always assume the vehicle's high voltage system is powered ON! Cutting, touching, crushing, or tampering with the vehicle's high voltage components can result in serious injury or death. Always use appropriate tools and use appropriate personal protective equipment (PPE). Failure to comply may result in serious personal injury or death.



WARNING: Electrical Shock Hazard. De-Energizing the high voltage (HV) system does not dissipate voltage inside the battery pack! HV battery packs remain energized. High voltage cables between battery packs and HV power distribution unit remain energized. Making contact with the high voltage battery pack internals or battery cables may result in serious personal injury or death.

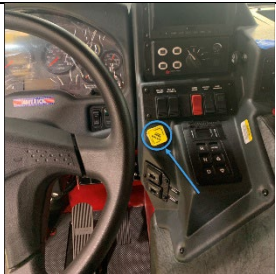


WARNING: Electrical Shock Hazard. Electric vehicles damaged in a crash may have compromised safety systems and may present a potential electrical shock hazard. Use caution and use proper protective equipment including high voltage safety gloves and boots. Remove all metallic jewelry. Isolate the high voltage system as directed. Failure to comply may result in serious personal injury or death.

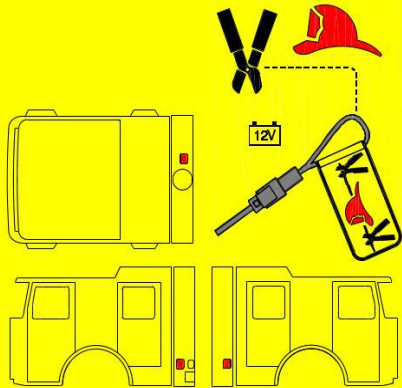


WARNING: Electrical Shock Hazard. Service and handling of high voltage components is to be restricted to qualified personnel only. Failure to follow this instruction and those applicable to your region may result in serious personal injury or death.

3.1 DISABLING PROCEDURE

	Engage Park Brake	Pull out to apply brake; push-in to release brake. Applying the parking brake automatically shifts the vehicle to neutral (N).	
	Key Off	Turn off ignition	
	Deactivate Low Battery via LV Battery Disconnect Switch	Turn low voltage disconnect to the off position to disconnect the low voltage battery.	

3.2 DISABLING THE HIGH VOLTAGE SYSTEM

	High Voltage Disconnect (Cut Loop)	Locate an accessible cut loop and cut a section out so the ends cannot touch 	ELECTRIC VEHICLE DRIVETRAIN IN THE EVENT OF AN EMERGENCY THE DRIVELINE POWER CAN BE DISCONNECTED BY BREAKING/SEVERING CUT LOOPS LOCATED IN ONE OF THREE LOCATIONS 
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3.3 DEACTIVATION IF VEHICLE IS CHARGING

If Charging 	Activate E-STOP if available on EVSE Wait for green light to turn off Unplug vehicle from charger Unlock Cab Flip low voltage disconnect switch below road side seat to off position	Charging Inlet Location (LS of Battery Rack) 
If Charge inlet is locked 	Activate E-STOP if available on EVSE Press release button on charger handle and attempt to unplug. Flip low voltage disconnect switch below road side seat to off position Mechanically unlock charging inlet if necessary	

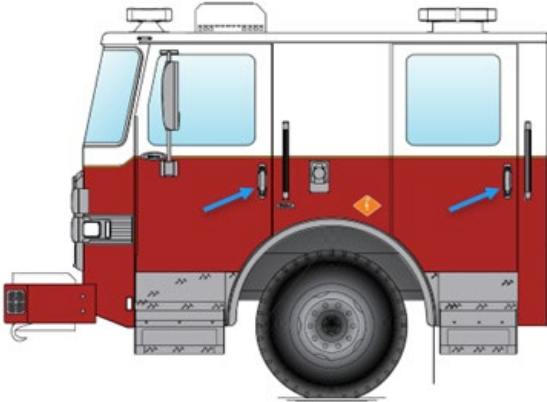
4. Access to the Occupants



WARNING: Pinch Point Hazard. Vehicle doors are heavy. Use caution when opening and avoid pinch points. Failure to comply may result in serious personal injury or death.



WARNING: Fall Hazard. Always maintain three (3) points of contact (such as one foot and two hands or one hand and two feet) when getting in or out of the vehicle. Use step and hand holds provided. Failure to comply may result in serious personal injury or death.

	Opening doors from outside		Pull door handle out towards you and pull door to open.
	Opening doors from inside		Pull door handle towards you and push out on door to open.



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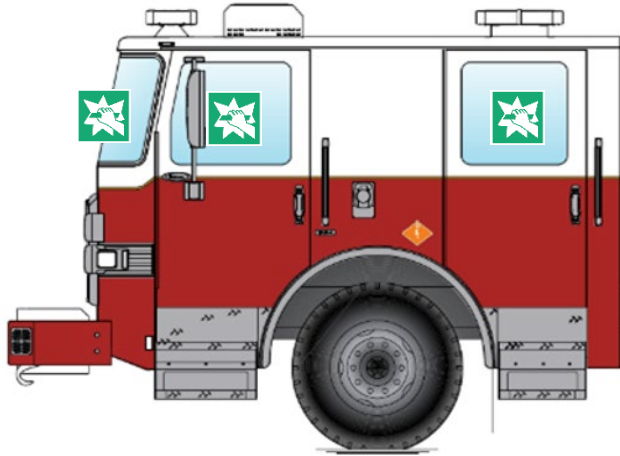
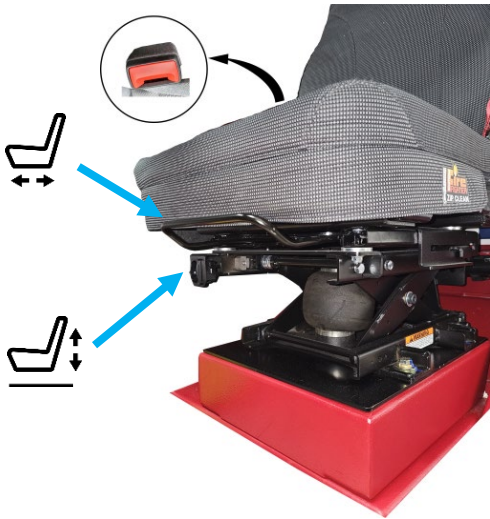
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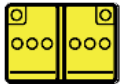
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	Break these windows to obtain access		Windshield is safety glass per FMVSS & ANSI.
Seat Adjustment and Seat Restraints To remove seat belt, push the red release button on the seat buckle.		 Height Adjust Air Control Switch Switch raises or lowers seat position.  Fore/Aft Slide Lift Bar Lift bar allows seat to slide fore/aft for desired position.	
Steering Column Adjustment		 Pull tilt/telescopic lever towards you to tilt steering column Push tilt/telescopic lever away to adjust steering wheel in or out.	

5. Stored Energy / Liquids / Gases / Solids

LIQUIDS/SOLIDS/GASES	Type	COLOR	HAZARDS ASSOCIATED
	700 V LI-ON	Colorless	     
	12 V Lead Acid	Colorless	     
Refrigerant 	R134A	Colorless	     
	Diesel Fuel	Colorless	  
Hydraulic Oil 	ISO 32	Dark Brown	  
Ethelene Glycol Coolant	50-50	Blue Green	  
Engine/Transmission Oil	TES 668	Red	  

6. In Case of Fire



WARNING: Inhalation Hazard. Fires in crash-damaged electric vehicles may emit toxic or combustible gasses. Wear personal protective equipment and self-contained breathing apparatus (SCBA) when working in close proximity or in a confined area. Ventilate the vehicle interior by opening the vehicle windows or doors. Ventilate the working area. Furthermore, take appropriate measures to protect civilians downwind from the incident. Failure to comply may result in serious personal injury or death.



WARNING: Explosion Hazard. Lithium-Ion batteries can explode during fire. Use caution and wear proper PPE. Failure to comply may result in serious personal injury or death.



Fire Hazard. Electric vehicles with damaged high voltage batteries require special handling precautions. Inspect the vehicle carefully for leaking battery fluids, sparks, flames, and gurgling or bubbling sounds. Contact emergency services immediately if any of these problems are observed. Failure to comply may result in a vehicle fire and serious personal injury or death.

6.1 FOR VEHICLE FIRES

Extinguishing Fire		Use Wet Foam when other materials are involved Use water to extinguish LI-ion Fires
Hazardous to Human Health		May Cause an allergic skin reaction Do not breathe dust, fumes, gas, mist, vapors, or spray Breather valves may emit large flames because of thermal runaway
Explosion Hazard		Explosive gas could accumulate – ventilate the working area. Move the vehicle outside buildings after extinguishing fire. Damaged LI-ion batteries could reignite; keep vehicle away from combustible materials
Corrosives		Causes skin burns and eye damage
High Voltage (700 V)		CATIII (1000 V) rated gloves required for exposed high voltage (HV) parts.
Battery Fire		May take 24 hours for battery pack to cool Remove sources of ignition Cool burned mass with water to cool battery cells Check LI- ion battery pack fires with thermal infrared camera (TIC or IR device)



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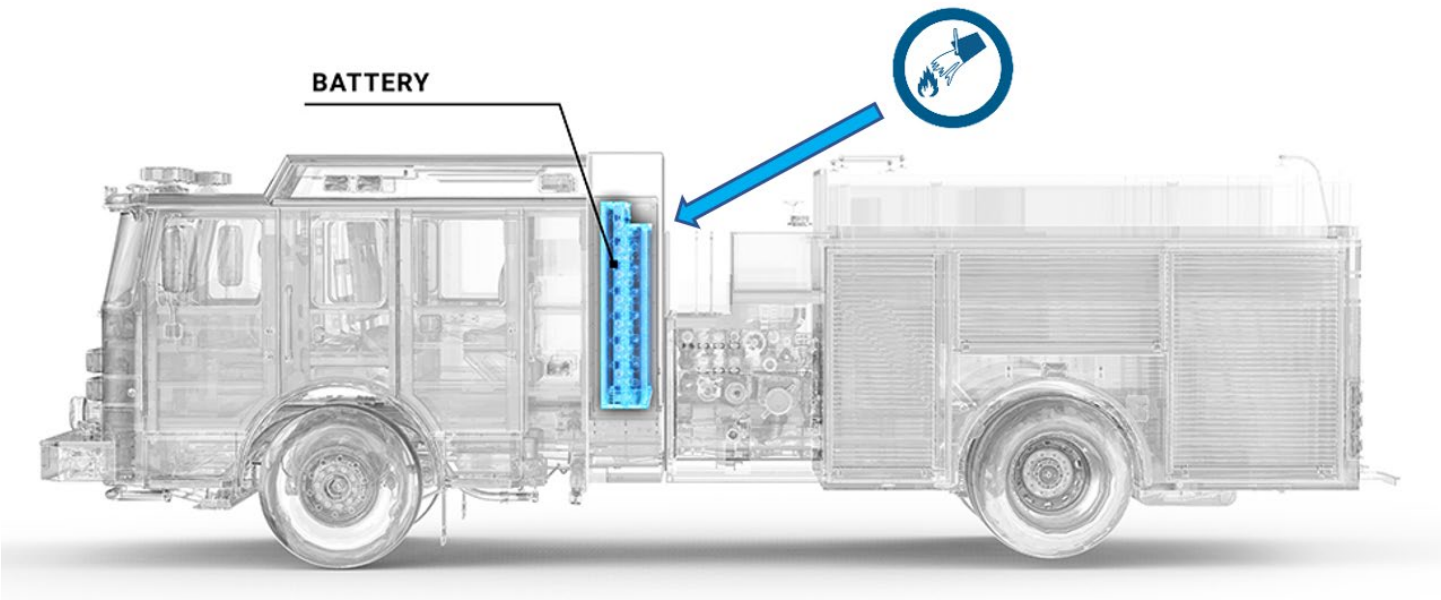
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6.2FOR BATTERY SYSTEM FIRES

Flood forward battery structure with water to cool structure.



7. In Case of Submersion



DANGER: Battery Failure Present Hazard. Visible bubbling or fizzing from a submerged or damaged electric vehicle may indicate high-voltage battery failure or an active electrical discharge. Do not touch, move, or attempt extraction of the vehicle while bubbling or fizzing is present. Maintain a safe distance until the reaction has stopped and the area is deemed safe. Failure to comply will result in serious personal injury or death.



WARNING: Electric Shock Hazard. High-voltage components and wiring may be energized and present a shock hazard. Always treat the high-voltage system as fully charged. Use appropriate PPE, including high-voltage safety gloves and boots, and remove all metallic objects such as jewelry before approaching or handling the vehicle. Failure to comply may result in serious personal injury or death.



WARNING: Electric Shock Hazard. A flooded vehicle that is plugged into a charging station can pose an electrocution hazard from the vehicle to the charging station. Always make sure that the vehicle is not plugged into a charging station before attempting to service. Failure to comply may result in a vehicle fire and serious personal injury or death.



WARNING: Fire Hazard. Flooded battery can still have stored energy and catch on fire.



WARNING: Electric Shock Hazard. Vehicles that have been submerged have a higher risk for a high voltage electrical battery fire and should be handled with more caution. The damage level of a submerged vehicle may not be visible. Submersion in water can damage the electrical components. It is recommended that fire fighters be present and ready in case of fire.

8. Towing / Transportation / Storage



WARNING: Electric Shock Hazard. Do not tow the vehicle until the rearmost driveshaft has been removed as this may cause the vehicle to generate electricity and can cause potential damage. Failure to comply may result in serious personal injury or death.



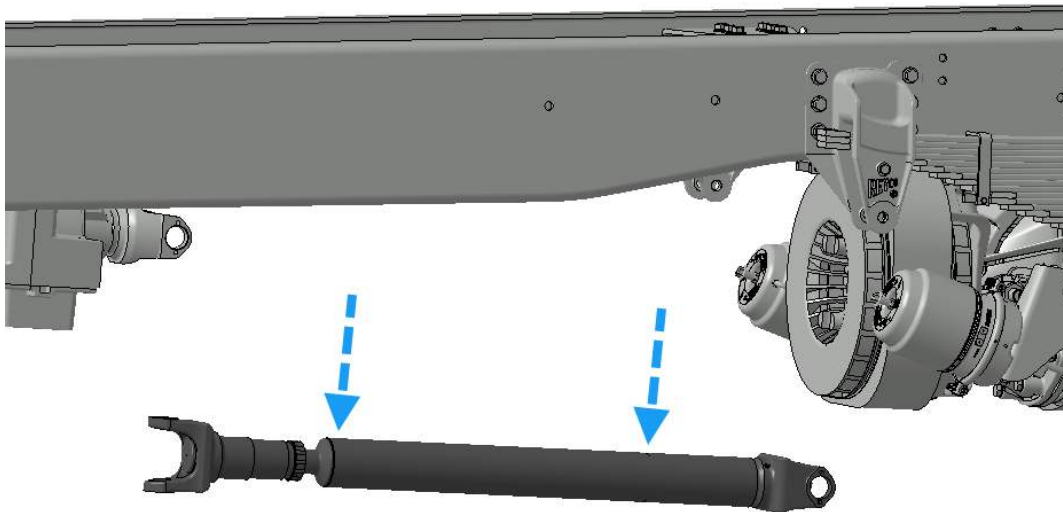
WARNING: Unintended Movement Hazard. Always approach the vehicle from the sides to stay out of the potential travel path. It may be difficult to determine if the vehicle is running due to lack of engine noise. Failure to comply may result in serious personal injury or death.



WARNING: Fire Hazard. Be alert. There is a potential for delayed fire with damaged lithium-ion batteries. Measure temperature with thermal camera before and after transportation. Failure to comply may result in a vehicle fire and serious personal injury or death.

8.1 TOWING/TRANSPORTATION PROCEDURE

- Do not tow the vehicle until the rearmost driveshaft has been removed as this may cause the vehicle to generate electricity.



- Drain all water from the water tank.
- Tow hooks and tow eyes may be found on the front or rear of the vehicle. The specific location will vary based on the options selected. These tow eyes and tow hooks may be used for towing disabled vehicles for short distances on flat and level surfaces in an emergency. They are not meant for long distance towing on public highways.



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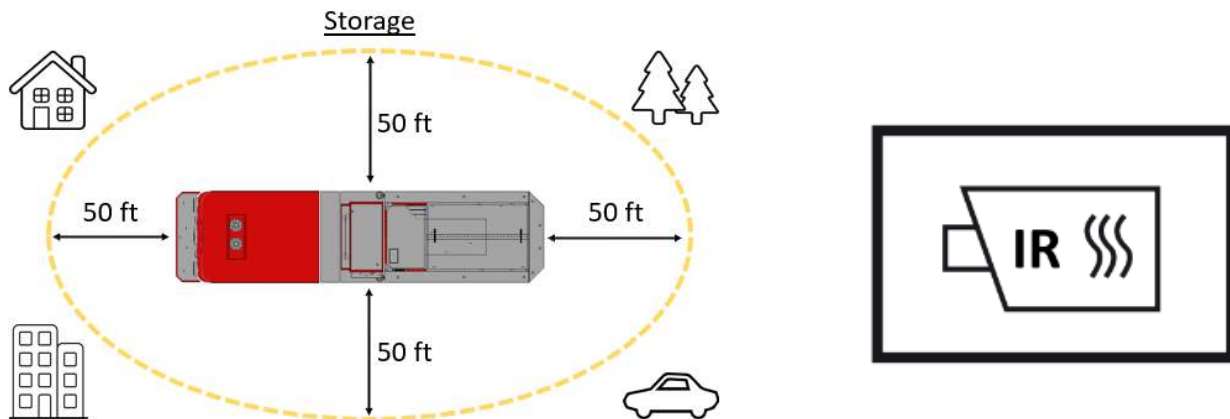
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Tire Lift Method

- Lift the front of the disabled vehicle using an underlift device with proper tire lift brackets that support and secure the front tires
- Do not lift the front of the disabled vehicle by attaching to or lifting by any component or structure of the front suspension.
- Before releasing the parking brakes, secure the connection to the towing vehicle.



8.2 STORAGE PROCEDURE



- Make sure the damaged vehicle is kept in an open area, not in an enclosed building.
- Inspect vehicle with thermal imaging equipment to ensure there are no hot spots before storing.
- Keep damaged vehicles away from surrounding vehicles, structures, and combustible items.
- If you see leaking fluids, sparks, smoke, flames, increased temperature, gurgling, popping or hissing noises from the high voltage battery compartment, evacuate the area and call 911.

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9. Important Additional Information

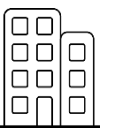


WARNING: Vehicle noise may be reduced in some operation modes. Failure to shutdown vehicle before immobilization could result in death, severe injury, or property damage.



WARNING: Lack of engine noise does not mean vehicle is off: Silent movement or instant restart capability exists until the vehicle is fully shut down.

10. Explanation of Pictograms Used

 Use water to extinguish fire	 Use ABC powder to extinguish fire	 Environmental Hazard	 Flammable	 Explosive	 Corrosive	 Hazardous to human health
 Acute toxicity	 Warning: High Voltage	 General Warning	 Warning: Low Temperature	 Thermal Infrared Camera	 Vehicle Charging Station	 Building
 House	 Car/Vehicles	 Trees/ Combustibles				